

An isolated character away from the scalar character

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_isolated_character_n
e_infinity`

theorem

Finds an isolated point in the open complement of the distinguished scalar character.

Statement

Let A be a complex C^* -algebra, with no unit assumed. On $\tilde{A} = \mathbb{C} \oplus A$, write $\chi_\infty(\lambda, a) = \lambda$. Let D be a norm-closed commutative unital $*$ -subalgebra of $\tilde{A}$, and assume its character space X is countable. Let $d \in D$ be nonzero and have scalar coordinate zero. Then there is $\chi \in X$ distinct from the scalar character $\chi_\infty|_D$ such that $\{\chi\}$ is open in X .

Assumptions

This theorem has no representation π or singleton-representation hypothesis. Countability of the character space is an explicit input.

Conclusion

The same character is both different from the distinguished scalar character and isolated in the full character space.

Proof route

Use the nonzero element to show that the complement of the scalar character is nonempty, apply the countable Baire lemma in that open subspace, and transfer openness back.

▼ Proof steps

1. If every character vanished on d , its Gelfand transform would be zero. Injectivity of the Gelfand transform would then give $d = 0$, a contradiction. Thus some χ_0 satisfies $\chi_0(d) \neq 0$. The scalar character has value zero at d by the given scalar-coordinate condition, so $\chi_0 \neq \chi_\infty|_D$.
2. Set $U = X \setminus \{\chi_\infty|_D\}$. The character space of the unital commutative C^* -algebra D is compact Hausdorff; hence the distinguished singleton is closed and U is open. Step 1 shows $U \neq \emptyset$. It is countable and T_1 as a subspace of X , and it is Baire, since an open subspace of this locally compact Hausdorff space is locally compact Hausdorff.

3. Apply An isolated point in a nonempty countable Baire space to this very subspace U , using the four properties just verified. It supplies $\chi \in U$ for which $\{\chi\}$ is open in U . The inclusion of an open subspace is an open map, so the same singleton is open in X . Membership in U is precisely $\chi \neq \chi_\infty|_D$.

Main citations

- Exact theorem and proof
- An isolated point in a nonempty countable Baire space
- Scalar-coordinate character

Lean source signature (exact)

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theorem exists_isolated_character_ne_infinity
  (D : StarSubalgebra ℂ (Unitization ℂ A))
  [IsClosed (D : Set (Unitization ℂ A))]
  [IsMulCommutative D]
  [Countable (WeakDual.characterSpace ℂ D)]
  (d : D) (hd : d ≠ 0)
  (hdfst : (d : Unitization ℂ A).fst = 0) :
  ∃ chi : WeakDual.characterSpace ℂ D,
    chi ≠ infinityCharacterOn (A := A) D ∧
    IsOpen ({chi} : Set (WeakDual.characterSpace ℂ D))
```

In the source	Mathematical meaning
$(D : \text{StarSubalgebra } \mathbb{C} (\text{Unitization } \mathbb{C} A))$	The unital complex $*$ -subalgebra D of the unitization of the ordered possibly non-unital A .
$[\text{IsClosed } (D : \text{Set } (\text{Unitization } \mathbb{C} A))]$	D is norm closed, hence a C^* -algebra in its induced norm.
$[\text{IsMulCommutative } D]$	Any two elements of D commute.
$[\text{Countable } (\text{WeakDual.characterSpace } \mathbb{C} D)]$	The full character space X of D is countable.
$(d : D) (hd : d \neq 0)$	A particular nonzero element d of D .
$(hdfst : (d : \text{Unitization } \mathbb{C} A).\text{fst} = 0)$	When d is viewed in $\tilde{A} = \mathbb{C} \oplus A$, its scalar coordinate is zero; equivalently $\chi_\infty(d) = 0$.
$\exists \text{ chi} : \text{WeakDual.characterSpace } \mathbb{C} D$	Choose one character χ of D .
$\text{chi} \neq \text{infinityCharacterOn } (A := A) D$	That χ differs from the scalar character restricted to D .
$\text{IsOpen } (\{\text{chi}\} : \text{Set } (\text{WeakDual.characterSpace } \mathbb{C} D))$	The singleton of the same χ is open in the full character space, not merely in an unspecified subset.

Exact source and proof.

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▼ **Exact content identity**

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